

Vivi Huang

Computer Engineering, University of Waterloo
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SKILLS

Languages: Python, Java, C/C++, TypeScript, JavaScript, SQL, ArkTS

Libraries/Frameworks: PyTorch, FastAPI, Spring, React, Vite, LangChain, OpenCV, MediaPipe, ONNX Runtime

Tools/Infrastructure: Git, Docker, CI/CD, Linux/Unix, Maven, PostgreSQL, Supabase, ChromaDB, Pytest, AWS

EDUCATION

•BASc, Computer Engineering

Sept 2023 - Apr 2028

University of Waterloo

GPA: 4.0/4.0

- Dean's Honour List (Fall 2023); Term Distinction in all semesters

EXPERIENCE

•Software Engineer Intern

Sept 2025 - Apr 2026

Huawei Technologies Canada, Co.

Markham, ON

- Owned a cycling-route extraction pipeline recovering ordered landmarks via **morphological processing** and route-snap via HuaweiMaps API. Cut runtime from **30s** → **13s** using **AsyncIO** for dual-pipeline architecture.
- Fine-tuned **YOLO26** for in-pipeline map icon detection on a self-curated **2.3k-image** dataset, a class absent from pretraining, reaching **98% mAP@50** with mosaic and flip augmentation
- Architected a **Windows desktop agent** utilizing **Win32gui** and a **ChromaDB-backed prefix tree** to index and persist workflow intent, replaying learned skills at **>80%** task success rate.
- Shipped a multi-device **HarmonyOS** live speech demo (**ArkTS/MVVM**) on **Pangu AI** + **LiveKit**, bridging LiveKit's JS SDK via ohos **ArkWeb**. Integrated internal audio models for real-time distance-based filtering
- Built full frontend for multi-person Xiaoyi voice agent gating replies on **voiceprint identity**, and cut p50 response latency from **7-8s to under 3s** by streaming **VAD/LLM/TTS** stages and speaker verification concurrently

•Backend Developer

January - April 2025

JD.com

Beijing, CN

- Contributed to one of JD.com's largest internal platforms (**Spring Boot**, **Redis**, **Kafka**), used by hundreds of marketing staff for coupon issuance and promotion eligibility across **500,000+ daily transactions**
- Implemented an internal promotion submission form; traced failing edge-cases and fixed front-end render defects using **React** with new alert abstraction

•Full Stack Developer

May 2024 - August 2024

Adaptive Pulse

Toronto, ON

- Introduced authentication to a B2B AI SaaS web app. Built a new **OAuth2 SSO** flow over Google Drive and Dropbox APIs so **100+ client companies** could bulk-import documents instead of uploading one file at a time
- Orchestrated continuous **ETL** sync from client sources into **Supabase/Postgres** with **Airbyte**, landing **1M+ rows** of client data
- Built a **RAG** summarization service over client tax filings and financial reports using **LangChain.js**, OpenAI embeddings, and **pgvector**, cutting document review time by **~30%** in internal benchmarking

PROJECTS

•Gestalt Engine: Real-time Gesture-Audio Interface [GitHub]

Feb 2026 - May 2026

- Implemented a stacked two-layer **LSTM** dynamic gesture recognizer twice — fully **from scratch in NumPy** (hand-written LSTM cell and backpropagation, **~85%** test accuracy) and in **PyTorch** (**~92%**) — across **21 dynamic gesture classes**, authoring a 500-line pipeline to transform **800k raw IPN-Hand frames** into training-ready arrays
- Served real-time inference at **<15ms per frame** in an **OpenCV** loop, and built the web app in **Vite**, **React**, and **FastAPI** on **MediaPipe Web SDK** landmarks, streaming predictions over websockets at **sub-100ms** end-to-end latency
- Prototyping a **spatial audio pipeline** to synthesize stereo output from 3D hand-tracking data, enabling **non-visual feedback** for gesture-based interfaces

•Anchor: Focus App & Extension — DeltaHacks 12 [Devpost]

Jan 2026

- Built the **Chrome extension** for a focus tool that detects when a user drifts off-task and intervenes with a spoken prompt and an on-screen overlay, using **JavaScript**, **Next.js**, and the **Chrome APIs** (team of 4)
- Streamed user activity into a **Gemini** pipeline with **ElevenLabs TTS** interrupts, cutting token usage by **75%** by **pHash**-deduplicating consecutive screenshots before inference
- Integrated an eye-tracking JS model into the RT tracking sessions to detect distractions